Catalog Export (Dictionary + Glossary + Lineage) and Impact Analysis Scenario

Project: MediLink Health Network Enterprise Data Platform Database platform: MySQL 8.0+ Data sources: Kampala, Mukono, Mbarara, and Gulu healthcare facilities

7.13 Data Catalog Export

7.13.1 Introduction

MediLink Health Network requires a central data catalog to document the structure, meaning, origin, transformation, and use of its healthcare data. The catalog will support data governance, improve consistency across healthcare facilities, and enable technical personnel and business users to understand how information is processed throughout the enterprise data platform.

The catalog export consists of three interconnected components: a data dictionary, a business glossary, and a data lineage register. Together, these components document the technical structure of data, the meaning of business terms, and the movement of data from source systems to analytical reports.

The catalog will initially be maintained using structured documentation and database metadata. As the platform develops, the registers can be stored in MySQL tables or exported to CSV or Excel for review and distribution.

7.13.2 Data Dictionary

The data dictionary documents the technical characteristics of the principal data elements used in the MediLink platform. It identifies their meanings, data types, validation requirements, and relationships.

The following table represents the proposed data dictionary. The entries must be reconciled with the final implemented MySQL schema before being treated as a verified database export

Table / Asset	Data element	Data type	Description	Validating / Governance rule
master_patient	enterprise_patient_id	VARCHAR	Unique MediLink patient identifier	Primary key or otherwise enforced as unique and non-null
master_patient	full_name	VARCHAR	Patient's recorded full name	Required according to approved registration rules; trim unnecessary spaces
master_patient	date_of_birth	DATE	Patient's date of birth	Must satisfy approved date validation rules
master_patient	Sex	VARCHAR	Recorded sex value	Must conform to the approved value set
master_patient	phone_number	VARCHAR	Patient contact number	Normalise valid Ugandan numbers to 256XXXXXXXXX
master_patient	national_id_hash	VARCHAR	Protected identifier used for authorised matching	Restrict access and validate format and provenance
patient_source_record	source_patient_id	VARCHAR	Patient identifier assigned by a source facility	Must be traceable to the originating system
patient_source_record	facility_code	VARCHAR	Code identifying the originating healthcare facility	Must reference an approved facility
patient_source_record	enterprise_patient_id	VARCHAR	Enterprise identifier linked to the source record	Must reference a valid master patient
fact_patient_visit	visit_key	BIGINT	Warehouse identifier for a visit fact	Primary key
fact_patient_visit	patient_key	BIGINT	Reference to the patient dimension	Must reference a valid dimension member
fact_patient_visit	facility_key	BIGINT	Reference to the facility dimension	Must reference a valid dimension member
fact_patient_visit	date_key	INT	Reference to the date dimension	Must reference a valid dimension member
fact_patient_visit	visit_count	INT	Measure representing a visit	Must follow the defined fact-table grain
etl_run_log	etl_run_id	VARCHAR	Identifier for an ETL execution	Must identify a specific processing run
etl_run_log	run_status	VARCHAR	Outcome of an ETL execution	Must use approved status values
etl_run_log	records_loaded	INT	Number of records successfully loaded	Must be a non-negative integer

7.13.3 Business Glossary

The business glossary defines important MediLink terms in language that healthcare personnel, management, data stewards, and technical staff can understand consistently.

Business term	Definition	Responsible owner	Approved usage
Master Patient	The centrally managed enterprise identity representing a patient across participating facilities	Data Governance Lead	Patient identity management
Golden Patient Record	The approved consolidated representation of a patient's verified attributes	Data Steward	Authorised patient lookup and integration
Enterprise Patient ID	The identifier used by MediLink to identify a patient across its integrated platform	Data Governance Lead	Cross-facility record linkage
Source Patient ID	The identifier assigned to a patient by an individual facility's source system	Facility Data Officer	Source record traceability
Patient Visit	A recorded healthcare encounter associated with a patient and a facility	Clinical Data Owner	Service utilisation and visit reporting
Data Quality	The degree to which data meets defined completeness, accuracy, validity, consistency, uniqueness, and timeliness requirements	Data Governance Lead	Quality assessment and monitoring
Data Steward	An authorised person responsible for maintaining data definitions, reviewing exceptions, and resolving data-quality issues	Data Governance Lead	Data correction and MDM review
Data Custodian	A person or team responsible for the technical handling, storage, and protection of data	IT Operations Lead	Database administration and operational controls
ETL	Extract, Transform, and Load; the process of extracting source data, transforming it, and loading it into a target system	Data Engineering Lead	Data integration and warehouse processing
Data Lineage	The traceable path of data from its source through transformations to its final destination or report	Data Engineering Lead	Auditing and impact analysis
Data Warehouse	A repository organised to support integrated reporting and analytical queries	IT Operations Lead	Business intelligence and reporting

Data Completeness	The proportion of eligible records containing the required data elements	Data Governance Lead	Data-quality scorecard
Phone-Number Completeness	The proportion of eligible patient records with a present, valid phone number	Data Governance Lead	Monitoring against the minimum 97% target
Ministry of Health Report	An approved report prepared for the relevant Ministry of Health reporting requirements	Reporting Manager	Monthly regulatory reporting
Slowly Changing Dimension	A dimensional modelling technique for managing changes to descriptive attributes over time	Data Engineering Lead	Historical warehouse analysis

The glossary will be maintained under change control. Changes to a definition that affects data processing or reporting must be communicated to the responsible data owners and technical teams.

7.13.4 Data Lineage Register

The data lineage register documents the movement and transformation of data from source systems to reporting assets. It enables MediLink to investigate discrepancies, trace data origins, and identify downstream dependencies when changes are proposed.

Source asset	Transformation / processing	Target asset	Downstream use
Kampala patient system	Extract records, validate fields, standardise values	Patient staging data	MDM and patient integration
Mukono patient system	Extract records, validate fields, standardise values	Patient staging data	MDM and patient integration
Mbarara patient system	Extract records, validate fields, standardise values	Patient staging data	MDM and patient integration
Gulu patient system	Extract records, validate fields, standardise values	Patient staging data	MDM and patient integration
Patient staging data	Match identifiers, assess duplicates, apply approved survivorship rules	`master_patient`	Enterprise patient lookup
Source patient records	Link facility identifiers to enterprise identities	`patient_source_record`	Cross-facility traceability
Visit staging data	Validate dates, map facility and service identifiers, resolve warehouse keys	`fact_patient_visit`	Visit analysis
Facility reference data	Standardise facility codes and descriptions	`dim_facility`	Facility comparisons
Date reference data	Generate calendar attributes	`dim_date`	Period-based reporting

Patient master data	Apply approved analytical attributes and privacy controls	`dim_patient`	Patient-level analysis where authorised
Warehouse fact and dimension tables	Aggregate visit counts by facility and period	Monthly visit report	Management and Ministry of Health reporting
ETL processing logs	Capture execution status, record counts, and errors	`etl_run_log`	Operational monitoring and audit

7.13.5 Catalog Maintenance and Export Procedure

MediLink will maintain the catalog through a controlled process:

1. Register each critical source, staging, master-data, and warehouse asset.
2. Extract technical information from MySQL's `INFORMATION_SCHEMA` where applicable.
3. Document business definitions, responsible owners, and validation requirements.
4. Record source-to-target mappings and the transformation rules used in ETL processes.
5. Update lineage records whenever a schema, transformation, or report changes.
6. Review the catalog periodically for completeness and accuracy.
7. Export the registers into a consistent CSV or Excel format for governance reviews and audit evidence.

The catalog will be versioned so that users can identify the definitions and transformation rules applicable to a particular reporting period.

7.14 Impact Analysis Scenario

7.14.1 Scenario: Change to the Patient Phone-Number Standard

MediLink currently receives patient phone numbers from four healthcare facilities. These numbers may be represented using different local formats, international prefixes, punctuation, or spacing.

Suppose MediLink approves a revised enterprise rule requiring all valid Ugandan patient phone numbers to be stored in the normalised format `256XXXXXXXXX`.

This change affects more than the patient database column. It may influence data ingestion, validation, MDM matching, warehouse attributes, quality indicators, and reports. An impact analysis is therefore required before implementation.

7.14.2 Business Reason for the Change

Standardising phone numbers will improve consistency across facilities, support reliable contact-data validation, reduce formatting-related matching problems, and improve the interpretation of the phone-number completeness indicator.

The change must not treat a common phone number as definitive proof that two records belong to the same patient. Family members may share phone numbers, and patient identity resolution must continue to use multiple attributes and appropriate verification procedures.

7.14.3 Impacted Data Assets and Processes

Component	Potential impact	Required action
Facility source systems	Existing records may use different formats	Document source formats and confirm extraction mappings
Patient staging tables	Existing validation may accept inconsistent formats	Update normalisation and validation rules
`master_patient`	Stored contact values may require standardisation	Validate existing values and update eligible records under controlled procedures
`patient_source_record`	Source references must remain traceable after corrections	Preserve original source identifiers and source values where required
MDM matching process	Phone-based matching scores may change	Review matching rules and test for false-positive and false-negative matches
`dim_patient`	Analytical contact attributes may need updating if retained	Update transformation logic and historical-handling rules
ETL jobs	Transformation and rejection logic may be affected	Update, test, version, and document ETL scripts
`v_dq_scorecard`	Phone completeness and validity calculations may change	Review metric definitions and validate results before and after deployment
Monthly reporting	Data-quality figures may change following standardisation	Confirm report logic and explain any discontinuities in historical trends
Metadata repository	Field definitions and validation documentation become outdated	Update the data dictionary, glossary, and lineage register
Audit and exception logs	New validation failures and corrections must be traceable	Record processing outcomes and correction reasons

7.14.4 Risk Assessment

The change presents the following risks:

- Data loss: Incorrect cleansing rules could remove digits or alter legitimate numbers.
- False patient matches: Matching rules based on phone numbers could incorrectly combine records belonging to different patients.
- ETL failures: Previously accepted values may be rejected by stricter validation.
- Misleading indicators: Phone-number completeness may appear to decline if the definition changes without explanation.
- Reporting inconsistencies: Historical reports may not be directly comparable with results generated under the revised rules.
- Loss of traceability: Updating source values without preserving the original references could make later investigation difficult.

These risks will be managed through testing, controlled deployment, data reconciliation, and documented approvals.

7.14.5 Implementation and Validation Plan

The proposed change will follow these steps:

Step 1: Document the change. Record the approved phone-number standard, its effective date, affected data assets, and responsible data owner.

Step 2: Identify dependencies. Use the data catalog and lineage register to identify staging tables, master-data rules, ETL jobs, warehouse dimensions, quality views, and reports that depend on phone-number values.

Step 3: Profile existing data. Measure missing, valid, invalid, and inconsistently formatted phone numbers in the current dataset. Preserve the baseline measurements.

Step 4: Develop and test transformations. Update the cleansing rules and test representative local-format, international-format, malformed, missing, and ambiguous values in a non-production environment.

Step 5: Test MDM matching. Verify that the revised standard improves matching consistency without automatically merging patients who share a phone number.

Step 6: Execute a controlled migration. Apply the approved transformation to eligible records, preserving source references and recording exceptions.

Step 7: Reconcile results. Compare records processed, successfully standardised, rejected, and left unresolved. Verify that patient counts and visit counts have not changed unexpectedly.

Step 8: Update metadata. Revise the data dictionary, glossary where necessary, lineage mappings, and relevant processing documentation.

Step 9: Approve deployment. Obtain approval from the responsible data owner and technical lead after successful testing and reconciliation.

Step 10: Monitor after deployment. Review phone-number completeness, validity, matching exceptions, and reporting outputs to identify unintended effects.

7.14.6 Success Criteria

The change will be considered successful when:

1. Eligible valid phone numbers conform to the approved format.
2. Invalid and ambiguous numbers are flagged rather than silently altered.
3. Original source references remain traceable.
4. MDM matching does not produce an unacceptable increase in incorrect merges.
5. The `v_dq_scorecard` accurately reflects the revised validation rules.
6. Affected ETL jobs and reports pass validation.
7. Data dictionary, glossary, and lineage documentation are updated.
8. The data owner approves the results and outstanding exceptions are documented.

MediLink will use actual pre-change and post-change measurements to assess success. No improvement percentage will be claimed until the transformation has been executed and its results verified.